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Models of Neuromodulation

Michael C. Avery¹ and Jeffrey L. Krichmar²¹ Systems Neurobiology Laboratory, Salk Institute for Biological Studies, La Jolla, CA, USA² Department of Cognitive Sciences, Department of Computer Science, University of California, Irvine, CA, USA

Introduction

Neuromodulatory systems signal important environmental events to the rest of the brain influencing cognitive behaviors, such as attention, decision making, learning, and memory. These neuromodulatory systems consist of small pools of neurons (on the order of thousands in the rodent and tens of thousands in the human) located in the brain stem, pontine nucleus, and basal forebrain (see Fig. 27.1). Ascending neuromodulatory systems include noradrenergic, serotonergic, dopaminergic, and cholinergic projections from the brain stem and basal forebrain regions to broad areas of the central nervous system, the thalamus, and hypothalamus (Briand, Gritton, Howe, Young, & Sarter, 2007). Neuromodulators track environmental signals, such as risks, rewards, novelty, effort, and social cooperation. In this chapter, we review computational models of neuromodulation. We specifically focus this review on the dopaminergic, serotonergic, cholinergic, and noradrenergic neuromodulators (see Fig. 27.1). These neuromodulators act directly on neurons through synaptic neurotransmitters, and there is a large body of empirical work on these neuromodulators that supports the computational models presented here.

Dopaminergic System

The ability to predict future rewards and learn from errors is fundamental to every decision we make and action we take in our lives. It has been demonstrated (Schultz, Dayan, & Montague, 1997) that dopaminergic neurons in the ventral tegmental area (VTA) and substantia nigra pars compacta (SNc) have responses that match quite well to the error signal that is key for adjusting our current predictions about the world. In particular, Schultz and colleagues (1997) found that when a reward is expected and it is received, there is no net change in dopaminergic responses (Fig. 27.2, middle). When a reward is given that is not expected, however, these neurons fire phasic bursts of action potentials (Fig. 27.2, top). These bursts have been hypothesized to reinforce behavior by activating the direct pathway (D1 receptors) of the basal ganglia. When a reward is expected and no reward is given, there is a small decrease in the response of these neurons (Fig. 27.2, bottom), strengthening inputs to the indirect pathway (D2 receptors) in order to inhibit this behavior. A recent model, which takes into consideration D1 and D2 receptors in the basal ganglia, explains how levodopa, a drug to treat Parkinson's disease enhances, but other dopamine agonists impair or have

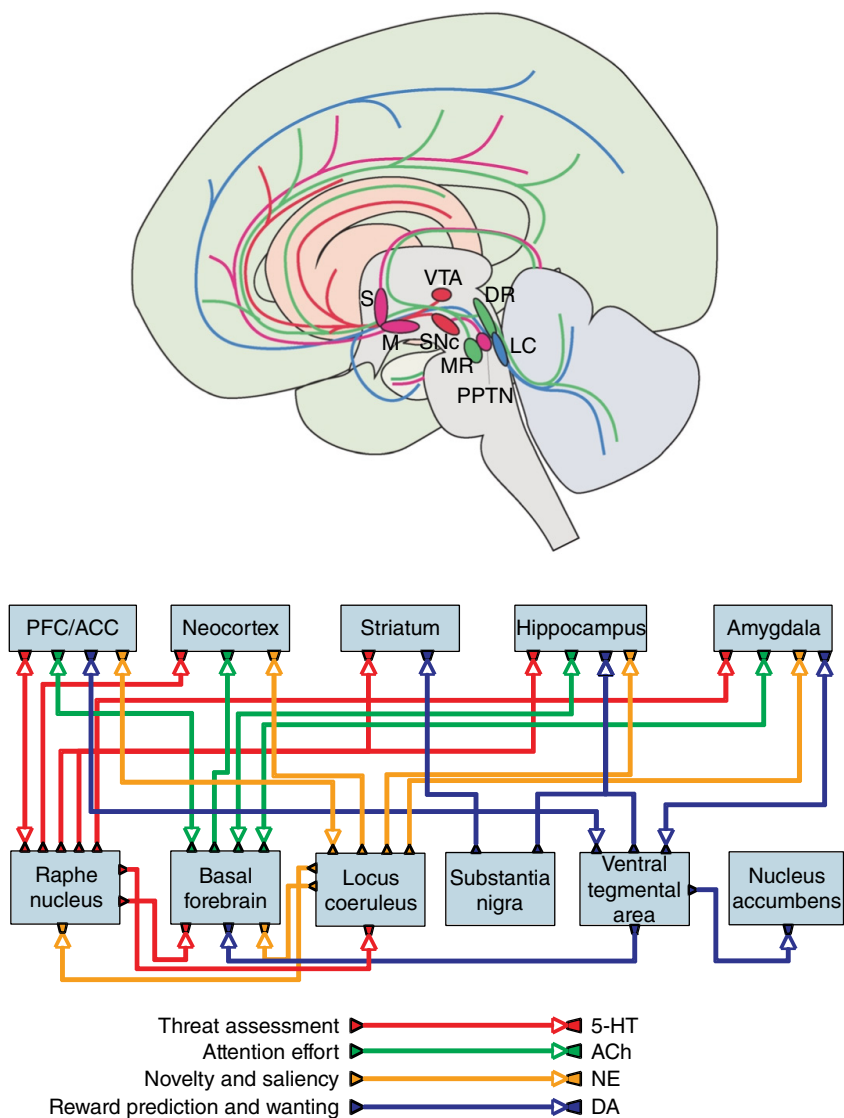


Figure 27.1 Sources and projections of the major neuromodulatory systems. (Left) The dopaminergic system is shown in blue, the serotonergic system in red, the noradrenergic system in orange, and the cholinergic system in green. From Doya (2002). (Right) Architecture of the neuromodulatory systems. From Krichmar (2008).

no effect on, stimulus–response learning and working memory (Moustafa, Herzallah, & Gluck, 2013).

In addition to the immediate response at the time of an unexpected reward, dopaminergic neurons also fire a phasic burst in response to a cue learned to be predictive of a reward. This is critical in that it demonstrates how the brain might associate the cue

as a secondary reinforcer. It suggests a means by which the brain is able to solve the credit assignment problem, which may occur in reinforcement learning as a result of the time difference between cue presentation and a reward. Furthermore, it suggests that the dopaminergic signal matches the temporal difference error signal seen in reinforcement learning, which will be discussed below, as

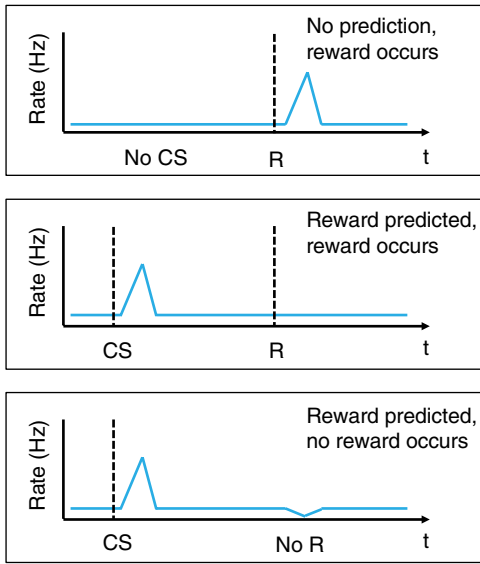


Figure 27.2 Responses of dopaminergic cells (top) before a stimulus is conditioned on the reward, (middle) after a stimulus is conditioned on the reward, and (bottom) when a reward is expected following a conditioned stimulus, but not given. From Schultz, Dayan, and Montague (1997).

opposed to a simpler algorithm such as the Rescorla–Wagner learning model (Rescorla & Wagner, 1972).

Reinforcement Learning and Dopamine

A machine learning method, known as reinforcement learning, has been used to impart the ability to learn and adapt behaviors on to artificial agents. Formally, the theory aims to find an optimal function, or policy, (P) for mapping states (S) into actions (A) that maximize the sum of future reward. Temporal difference learning methods, such as the actor–critic model, solve this problem by computing a reward prediction error signal, which is used in the updating of a value function (reward expectation) and policy as shown in the equations below.

$$\begin{aligned}\delta_t &= r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \\ V(s_{t+1}) &= V(s_t) + \alpha * \delta \\ P(a | s_{t+1}) &= P(a | s_t) + \alpha * \delta\end{aligned}\quad (\text{Eq. 27.1})$$

where r_{t+1} is the observed reward at time $t+1$, $V(s_t)$ is the value of state S at time t , gamma, γ , is a discounting factor, and α is the learning rate. The algorithm works by sampling the environment, making predictions, and then adjusting the predictions based on the error signal as shown in the feedback loop in Fig. 27.3. The ability to use the prediction error signal to update value estimates and behavioral policies is what gives this algorithm (and organisms) the flexibility to adapt to a dynamic environment. Recent models have extended this to other neuromodulatory systems and have attributed the α parameter, which controls the rate of learning, to the serotonergic (Balasubramani, Chakravarthy, Ravindran, & Moustafa, 2015) or noradrenergic systems (Nassar et al., 2012).

Circuits Involving Dopamine

The VTA and SNc are strongly influenced by the striatum and subcortical structures such as the lateral habenula and pedunclopontine tegmental nucleus. It has been shown that the phasic increase and dip in dopamine responses are due to the activation of the pedunclopontine tegmental nucleus and lateral habenula, respectively (Hong & Hikosaka, 2014; Matsumoto &

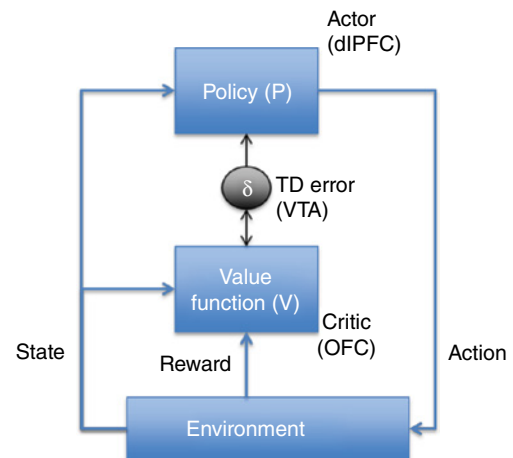


Figure 27.3 Actor–critic model including examples of brain regions that might be involved in computing each variable.

Hikosaka, 2007). Phasic increases also may be due to collicular or other sensory or nonsensory inputs to VTA/SNc when a salient event is identified (Redgrave & Gurney, 2006). Dopamine neurons, in turn, send projections to the striatum, thalamus, amygdala, hippocampus, and prefrontal cortex, demonstrating the “feedback” nature of this circuit.

We recently developed a model showing that activation of D2 receptors in the prefrontal cortex could influence working memory through interactions with the striatum (Avery & Krichmar, 2015). This model was based on the hypothesis that corticostriatal loops are involved in the gating of working memory (Frank, Loughry, & O'Reilly, 2001). Given that D2 receptors regulate the activity of neurons projecting to the striatum, they should regulate how and when this gate was opened or closed. The model demonstrates that improper activation of D2 receptors in the prefrontal cortex not only could lead to excess noise in prefrontal cortex, which would impair working memory, it could also lead to improper reward processing. Given this, we suggested that D2 receptors in the prefrontal cortex could play a large role in producing many of the symptoms of schizophrenia.

Serotonergic System

Serotonergic projections, which originate in the raphe nuclei of the brain stem, extend to almost all forebrain areas (Barnes & Sharp, 1999). The expression of serotonin or 5-HT occurs throughout the cortex, ventral striatum, hippocampus, and amygdala (Harvey, 2003; Meneses & Perez-Garcia, 2007). The raphe receives strong connections from the prefrontal cortex and the anterior cingulate cortex (Briand et al., 2007). Serotonin influences a broad range of decision-based functions such as reward assessment, cost assessment, impulsivity, harm aversion, and anxious states (Asher, Craig, Zaldivar, Brewer, & Krichmar, 2013).

Serotonin and Impulsivity

Several studies have investigated serotonin's involvement in reward and impulsivity, which is the tradeoff between taking an immediate reward, or else waiting for a future reward. In the temporal difference learning rule, this term is called temporal discounting or gamma (γ) in the TD equation (see Eq. 27.1). Kenji Doya suggested that serotonin levels may be related to temporal discounting (Doya, 2002). His group has confirmed this prediction in rodent and human experiments (Miyazaki, Miyazaki, & Doya, 2011; Tanaka et al., 2007). In addition, it has been shown that forebrain serotonin depletion leads to premature actions and steepens discounting of delayed rewards (Winstanley, Dalley, Theobald, & Robbins, 2003). In a rodent reward choice study, Doya and colleagues showed that increased serotonin neuron firing facilitates a rat's waiting behavior in prospect of forthcoming rewards and that higher serotonin activation enabled longer waiting, as would be predicted by temporal discounting (Miyazaki et al., 2011). In their task, there was a tone stimulus delay and a reward delay. The rat could choose whether to wait for a delayed reward or to quit the task. The wait errors associated with lower 5-HT neuron firing suggest that 5-HT can affect choice involving delayed rewards. The link between serotonin and temporal discounting has been explored using the acute tryptophan depletion (ATD) procedure. Lowering 5-HT levels via ATD influences a subject's ability to resist a small immediate reward over a larger delayed reward (Schweighofer et al., 2008; Tanaka et al., 2007; Tanaka et al., 2009). As such, subjects that underwent ATD had both an attenuated assessment of delayed reward and a bias toward small immediate reward, which were indicative of impulsive behavior and higher temporal discounting.

Serotonin and Harm Aversion

Serotonin (5-HT) has been linked to predicting punishment or harm aversion (Cools, Roberts, & Robbins, 2008; Crockett,

Clark, Apergis-Schoute, Morein-Zamir, & Robbins, 2012; Crockett, Clark, Tabibnia, Lieberman, & Robbins, 2008; Seymour, Daw, Roiser, Dayan, & Dolan, 2012). ATD caused subjects to be aggressive and risk taking by rejecting more monetary offers in the Ultimatum Game (Crockett et al., 2008). In a related study, Crockett and colleagues utilized the ATD procedure with a Go/No-Go task to show that lowering 5-HT levels resulted in a decrease in punishment-induced inhibition (Crockett, Clark, & Robbins, 2009). In a follow-up study, they investigated the mechanisms through which 5-HT regulated punishment-induced inhibition by using the ATD procedure paired with their Reinforced Categorization task, a variation on the Go/No-Go task (Crockett et al., 2012). Together, these results suggest that 5-HT influences the ability to inhibit actions that predict punishment and to avoid harmful circumstances.

Using an actor–critic model, (Asher, Zaldivar, & Krichmar, 2010; Zaldivar, Asher, & Krichmar, 2010) constructed a neural network where a reward critic represented the dopaminergic system and a cost critic represented the serotonergic system (see Fig. 27.4). In these experiments, the actor–critic agent played the socioeconomic game of hawk–dove against agents with fixed strategies. In the hawk–dove game, players choose between two actions: escalate (an aggressive tactic to seize a resource) or display (a cooperative tactic to share a resource). If both players escalate, they fight, resulting in an injury or penalty, which could either be serious or mild. If only one player chooses to escalate, then the escalating player receives the total value of the resource, and the other player receives nothing. If both players choose to display, then the value of the resource is shared. The reward critic tracked the expected value of the resource, and the cost critic tracked the expected punishment from an injury.

The simulations showed that the model was sensitive to the other player's strategy and the game environment (i.e., the likelihood of receiving a serious injury). The simulations predicted that impairment of the serotonergic system would lead to perseverant, uncooperative behavior. A simulated lesion of the serotonergic system resulted in the agent almost always engaging in risk taking (or lack of harm aversion) behavior, which was similar to behavior seen in human studies where serotonin levels were lowered via ATD while subjects played games such as Prisoner's Dilemma and the Ultimatum game (Crockett et al., 2008; Wood, Rilling, Sanfey, Bhagwagar, & Rogers, 2006).

Serotonin and Anxiety

Beyond punishment and impulsivity, 5-HT has been implicated in stress and anxiety (Jasinska, Lowry, & Burmeister, 2012; Millan, 2003). A review by Jasinska and colleagues proposed a mechanistic model between environmental impact factors and genetic variation of the serotonin transporter (5-HTTLPR), linking to the risk of depression in humans (Jasinska et al., 2012). They argued that genetic variation might be linked to a balance in the brain's circuitry underlying stressor reactivity and emotion regulation triggered by a stressful event, ultimately leading to depression. A review by Millan described studies showing that 5-HT function has been tied to an organism's anxious states triggered by conditioned or unconditioned fear (Millan, 2003). Manipulations to 5-HT_{1A} and 5-HT_{2A} receptors in the mouse resulted in elevated anxiety in the open-field test as measured by inhibition of center locomotion, and inhibition of responses to a novel object (Heisler et al., 1998; Weisstaub et al., 2006). Together, this work suggests a functional role for 5-HT in the control of anxious states.

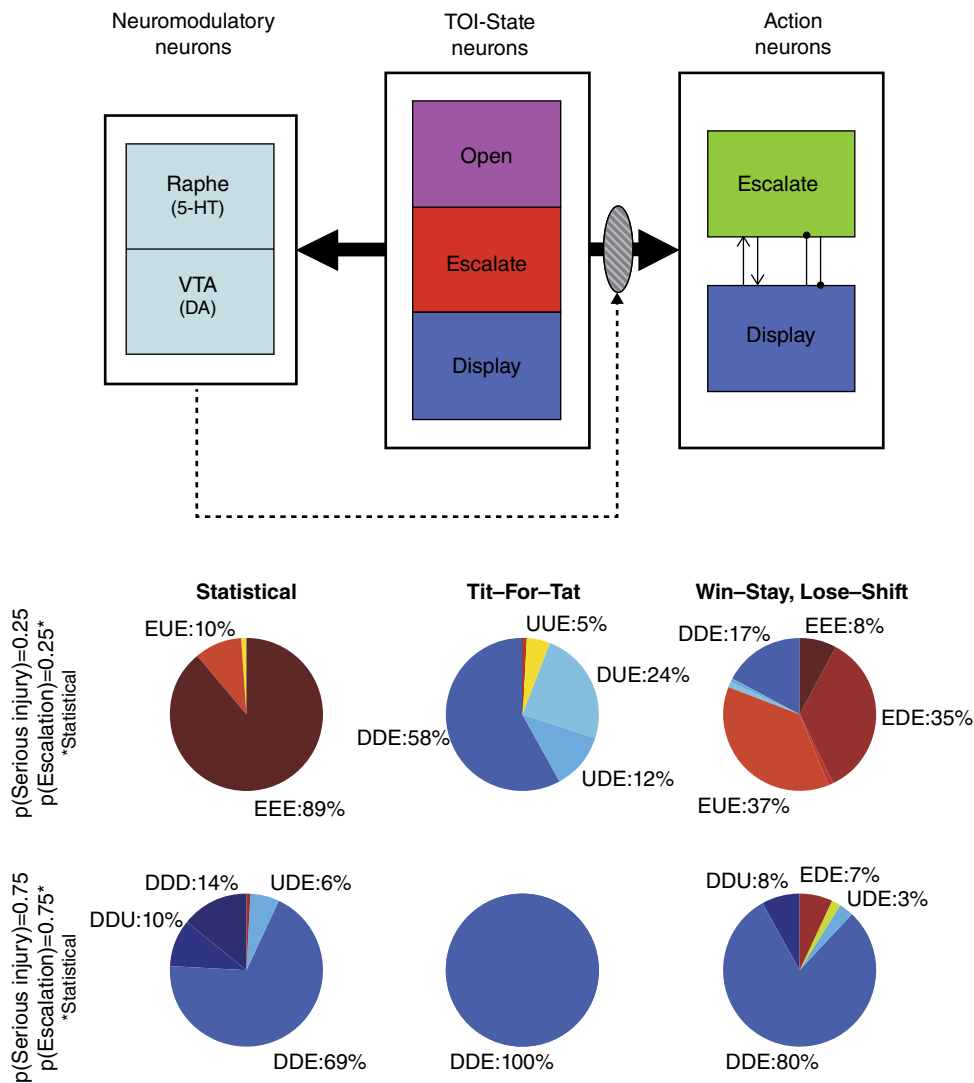


Figure 27.4 Neuromodulation effects in simulation of the hawk–dove game. (Top) Architecture of the neural model (neuromodulatory: Raphe and VTA; TOI-State: open, escalate, and display; and two action: escalate and display). The solid arrows extending from the TOI-State neurons represent all-to-all connections. The shaded oval and dotted arrows represent plasticity of the pathways. Within the action neurons, the line with the arrow at the end represent excitation, and the line with the dot at the end represent inhibition. (Bottom) The pie charts show the proportion of probable actions taken by the neural agent over 100 games. There are three states (open, escalate, and display), and three outcomes the neural agent can commit to: escalate (E), display (D), or undecided (U). Undecided represents random choice between “E” and “D.” The labels represent the neural agent’s response to the three TOI-State areas. Strategies that are dove-like are displayed in blue, hawk-like are displayed in red, and arbitrary strategies displayed in yellow.

Dopamine and Serotonin
Opponency

A theoretical review by Boureau and Dayan suggested that the serotonergic and

dopaminergic systems often activate in opposition for goal-directed actions (Boureau & Dayan, 2011). Opponency between the serotonergic system and the dopaminergic system has been proposed behaviorally

and in theoretical models (Daw, Kakade, & Dayan, 2002; Tops, Russo, Boksem, & Tucker, 2009). In this notion, dopamine triggers invigorated, reward-seeking behavior, and serotonin triggers withdrawn and punishment avoiding behavior. One model that investigated these opponent interactions, suggested that tonic serotonin tracked the average reward rate and that tonic dopamine tracked the average punishment rate in a similar context, and speculated that a phasic serotonin signal might report an ongoing prediction error for future punishment (Daw et al., 2002). However, it has been difficult to find empirical evidence supporting these roles for tonic and phasic neuromodulation. Modeling has shown that direct opponency between these systems is not necessary to achieve behavioral opponency (Asher et al., 2010; Zaldivar et al., 2010). In many cases there is an environmental tradeoff between the expected rewards and costs, and this can lead to opponency between active reward-seeking and withdrawn behavior. Indeed, by having different neuromodulatory systems handle different sensory events, this type of opponency emerges in these simulations.

A neurorobot model took into consideration the notion that the dopaminergic and serotonergic systems are in opposition by specifically having the serotonergic system inhibit the dopaminergic system (Krichmar, 2013). The algorithm was implemented in a neural network that controlled the behavior of an autonomous robot in the open-field paradigm (see Fig. 27.5). The opponency resulted in sensory events leading to withdrawn anxious behavior, such as wall following and finding its nest (i.e., the robot's charging station) when serotonin levels were high, and risky, reward-seeking behavior, such as moving to the center of the arena or investigating a novel object when dopamine levels were high. Furthermore, the algorithm tested the idea that top-down signals from the frontal cortex to neuromodulatory areas are critical for an organism to cope with both stressful and novel events. When the model's medial prefrontal cortex was lesioned, the robot withdrew to the outer

wall or its charging station after a stressor such as a bright light or collision. The model further suggested that projections from the orbitofrontal cortex to the dopaminergic ventral tegmental area (VTA) have a similar function when responding to a positive valence event. When the orbitofrontal cortex was artificially lesioned, the robot obsessively explored the center of the room and objects in the room.

Cholinergic System

The cholinergic system originates in the substantia innominata/nucleus basalis region (SI/nBM) and medial septum/vertical limb of the diagonal band (MS/VDB), which together comprise the basal forebrain. In particular, the SI/nBM has been shown to influence the neocortex and the MS/VDB has been shown to influence the hippocampus and cingulate especially in tasks involving attention.

To test the effect of the different contributions of the ACh projections from the SI/nBM and the MS/VDB on incrementing and decrementing attention, Oros and colleagues developed a neural simulation and tested it in a variety of behavioral paradigms that require attention (Oros, Chiba, Nitz, & Krichmar, 2014). The model exhibited behavioral effects such as associative learning, latent inhibition, and persisting behavior (see Fig. 27.6) and demonstrated that the ACh decremental pathway originating in the MS/VDB is necessary for appropriate learning and attention under dynamic circumstances and suggests a canonical neural architecture for decrementing attention.

Attention is also strongly modulated by acetylcholine through its projections to sensory cortices (Sarter, Givens, & Bruno, 2001; Sarter, Hasselmo, Bruno, & Givens, 2005). Deco and Thiele (Deco & Thiele, 2011) developed a spiking neural network model that suggested several important mechanisms that mediate the enhancement of top-down attention. The model incorporated several key cellular and synaptic changes resulting from cholinergic modulation

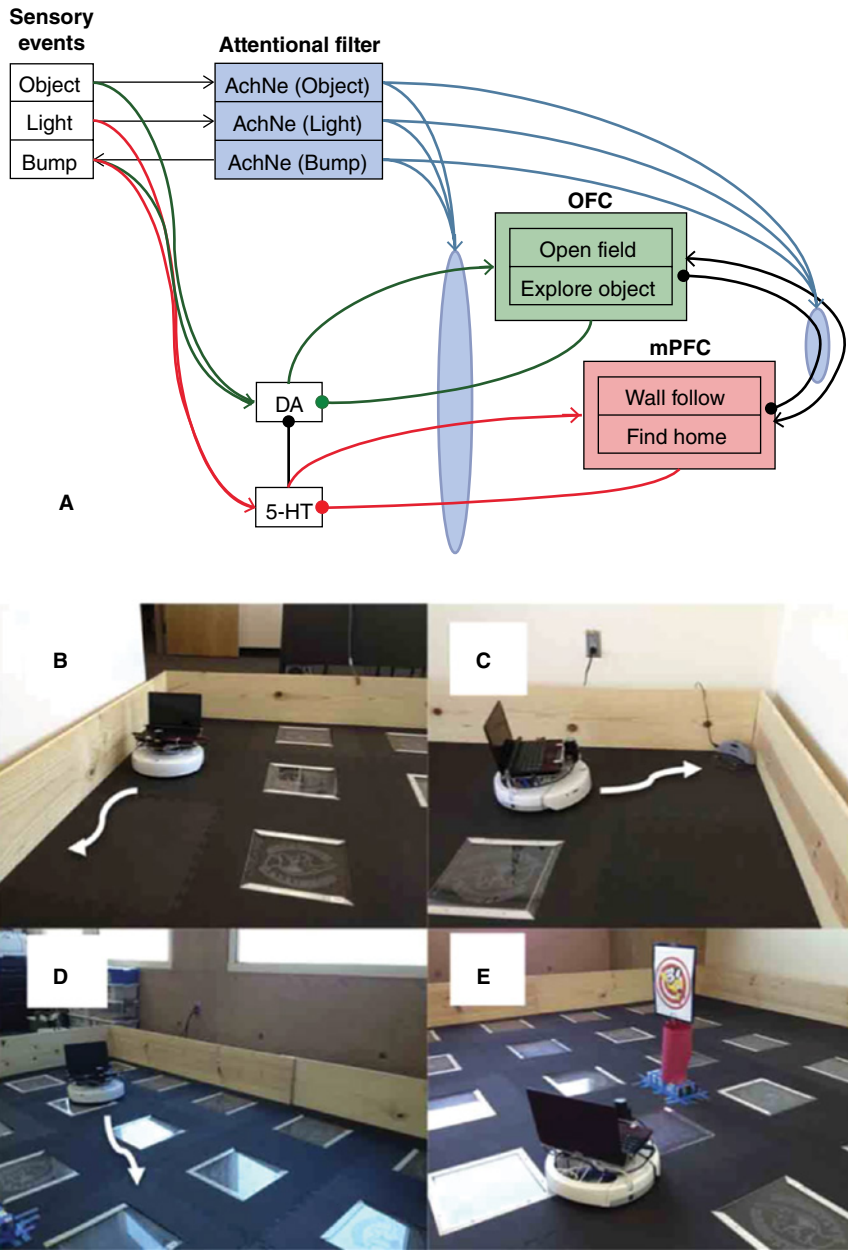


Figure 27.5 Embodied model of neuromodulation in an open-field test experiment. (A) Neural model architecture. (B) Wall following behavior. (C) Find home behavior. Finding home consisted of locating the robot's docking station. (D) Open-field behavior. The robot moved toward open spaces in the environment based on laser range finder readings. (E) Explore object. The robot approached narrow objects based on laser range finder readings. From Krichmar (2013).

including reduction in firing rate adaptation, enhanced thalamocortical input, reduction in lateral connectivity strength, and an increase in inhibitory drive. They suggest

that muscarinic enhancement of attention is mediated by suppression of intracortical connections and an increase in inhibitory drive. Again, this highlights the importance

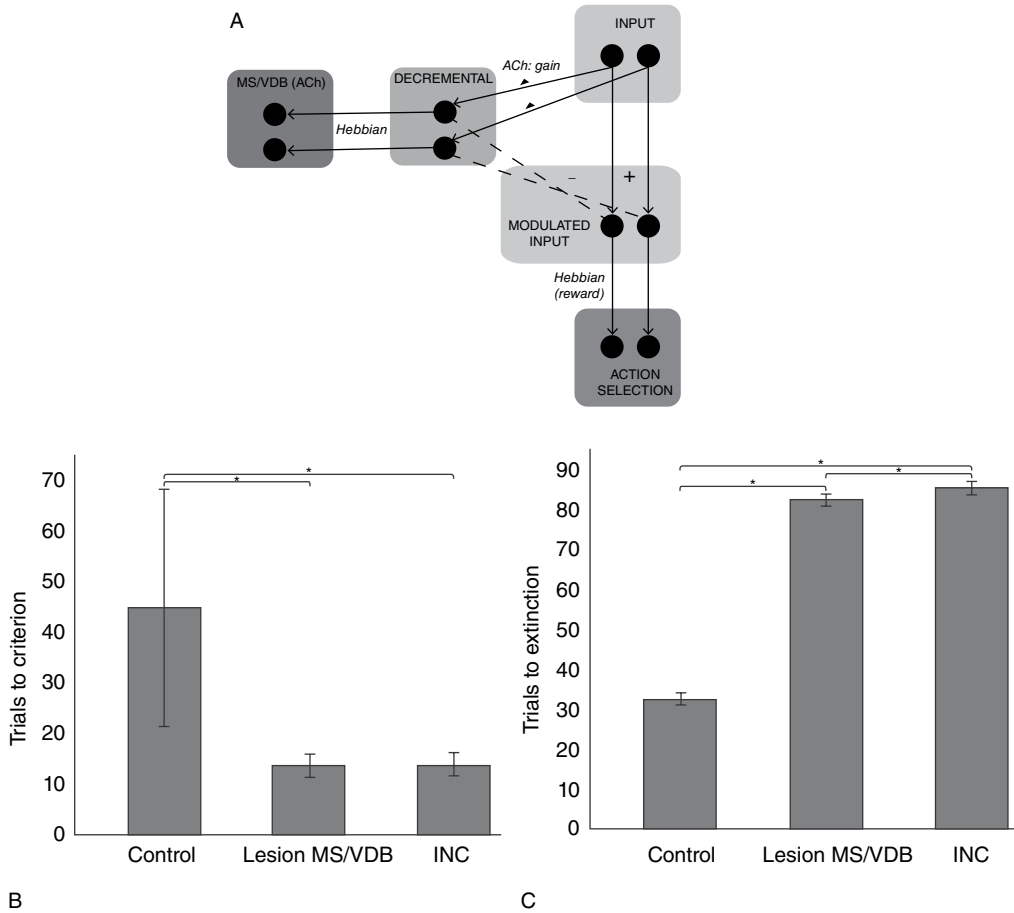


Figure 27.6 Model of acetylcholine's effect on decremting attention. (A) Neural architecture consisting of an input area, a decremental area, a cholinergic neuromodulatory system (MS/VDB), a modulated input area, and an action selection area. (B) Latent inhibition task. The intact model exhibited latent inhibition. Lesion of the MS/VDB disrupted latent inhibition. (C) Extinction task. Lesion of the MS/VDB during the extinction caused a significant increase in perseverative behavior. From Oros, Chiba, Nitz, and Krichmar (2014).

of acetylcholine in suppressing a very specific set of connections (intracortical) and potentially enhancing a broader class of behaviorally relevant inputs, which may include emotional, cognitive, or memory.

We recently developed a model (Avery, Dutt, & Krichmar, 2014) that took a slightly different approach and suggested that local and global activation of the cholinergic system might account for goal-directed attention and sensory-driven attention, respectively. In this model, stimulation of the basal forebrain has a global effect on the brain and enhances sensory input by disinhibiting the sensory thalamus via inhibitory projections from the basal

forebrain to the thalamic reticular nucleus. The model dissociates this enhancement of sensory input from the cholinergic enhancement of top-down input, which suggests that sensory enhancement is mediated by a local release of acetylcholine and activation of muscarinic receptors on inhibitory neurons. Similar to the Deco and Thiele model, this model stresses the importance of muscarinic receptors on inhibitory neurons. The model demonstrates that activation of muscarinic receptors is primarily involved in reducing noise correlations between neurons, which have been shown to influence information processing capabilities in the cortex.

Noradrenergic System

Noradrenergic neurons originate in the locus coeruleus (LC) and project to nearly every cortical and subcortical region except for the basal ganglia (Berridge & Waterhouse, 2003). The LC receives inputs from brain stem structures, but is also highly regulated by the prefrontal cortex, highlighting its role in integrating low-level autonomic and cognitive information and broadcasting this signal throughout the brain. Traditionally the noradrenergic system was thought to mediate arousal levels through slow changes in tonic levels of activation. Phasic activation of the LC, however, characterized by short bursts of activity, has taken on an important role in behavioral adaptation and task performance (Aston-Jones & Cohen, 2005; Aston-Jones, Rajkowski, Kubiak, & Alexinsky, 1994).

Phasic activation of the LC typically occurs in response to salient or novel inputs (Sara, Dyon-Laurent, & Herve, 1995; Vankov, Herve-Minvielle, & Sara, 1995) as well as task-relevant conditioned stimuli. If a reward is not associated with the novel stimulus, the response will eventually attenuate, which is likely important for transitions between phasic and tonic states. Task performance is optimal when LC neurons can be phasically activated and declines with increasing or decreasing tonic activity. Therefore, an inverted-U relationship between tonic LC activity and task performance exists that resembles the Yerkes–Dodson relationship between arousal levels and task performance (Fig. 27.7). This inverted-U nature of noradrenergic function in terms of signal detection and task performance has also been shown in working memory in the prefrontal cortex (Avery, Dutt, & Krichmar, 2013; Vijayraghavan, Wang, Birnbaum, Williams, & Arnsten, 2007; Wang et al., 2007). That is, too little or too much noradrenaline will likely impair working memory, but could also contribute to cognitive flexibility.

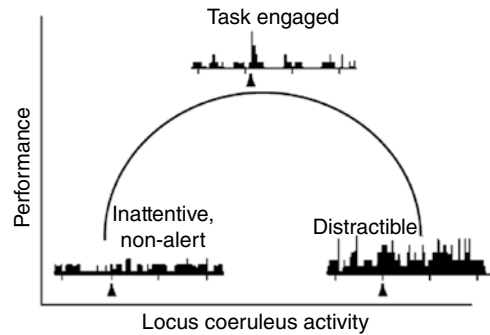


Figure 27.7 Inverted-U relationship between tonic LC activity and task performance.

Network Reset and Uncertainty

The noradrenergic system responds strongly to unexpected changes in the environment as well as task-relevant stimuli, which signal a change in behavior. This has led researchers to hypothesize that phasic activation of the LC is important for a “network reset” that induces a large-scale reconfiguration of neuronal activity across the brain to allow for changes in behavior and cognition (Bouret & Sara, 2005). It has also been shown that stress, which directly involves the noradrenergic system, can similarly induce a large-scale reconfiguration of functional activity in the brain and that the reconfiguration is dampened when subjects are given a drug to block adrenergic receptors (Hermans et al., 2011).

Yu and Dayan (2005) developed a Bayesian model based on the noradrenergic system’s involvement in cognitive shifts and response to novelty. Their model suggested that phasic bursts of LC activity encode unexpected uncertainty, which can be thought of as a large change in the environment that evokes a “surprise” response. Unexpected uncertainty acts to inform the model that a significant change has happened and the priors need to be updated. We developed a neural network model in order to understand the neural mechanism by which this computation might be implemented (Avery, Nitz, Chiba, & Krichmar, 2012). Our model

suggests that updating of priors was facilitated by a noradrenergic surprise signal that modulated the learning rate of Hebbian connections as well as the strength of sensory inputs.

Universal Models of Neuromodulation

Most models of neuromodulation focus on one or sometimes two neuromodulators. However, there is strong interaction between all of these systems. An exploratory survey of cholinergic, dopaminergic, noradrenergic, and serotonergic receptor expression using the Allen Mouse Brain Atlas showed that the substantia innominata of the basal forebrain, which is a source of cholinergic innervation, and the ventral tegmental area, which is a source of dopaminergic innervation, displayed high receptor expression of all four neuromodulators (Zaldivar & Krichmar, 2013). Since the nuclei of these neuromodulatory systems are thought to be the source of specific neurotransmitters, the projections from these nuclei to target regions may be inferred by receptor expression and suggest that neuromodulatory systems are highly interactive.

Neuromodulation and Temporal Difference Learning

Based on a review of experimental data and theoretical models, Doya proposed that the neuromodulatory system reflected the parameters found in the temporal difference learning rule (Doya, 2002). Eqs. 27.2–27.4 show the standard TD learning rule and Softmax function. Dopamine signals the error in reward prediction (δ in Eq. 27.2), serotonin controls the discounting of reward prediction (γ in Eq. 27.2), acetylcholine controls the speed of memory update (α in Eq. 27.3), and noradrenaline controls the randomness (i.e., the exploration/exploitation tradeoff) in action selection (β in Eq. 27.4). The possible interactions between those

neuromodulators and the environment are predicted on the basis of a computational theory of metalearning.

$$\delta(t) = r(t) + \gamma V(s(t)) - V(s(t-1)) \quad (\text{Eq. 27.2})$$

$$\Delta w = \alpha \delta(t) V(s(t)) \quad (\text{Eq. 27.3})$$

$$p(a) = \frac{\exp(\beta s_a(t))}{\sum_i \exp(\beta s_i(t))} \quad (\text{Eq. 27.4})$$

As has been discussed above, there is much evidence supporting dopamine and reward prediction, serotonin and temporal discounting. Experimental evidence supports the locus coeruleus being involved in the exploration/exploitation tradeoff, which is controlled by β in Eq. 27.4 (Aston-Jones & Cohen, 2005).

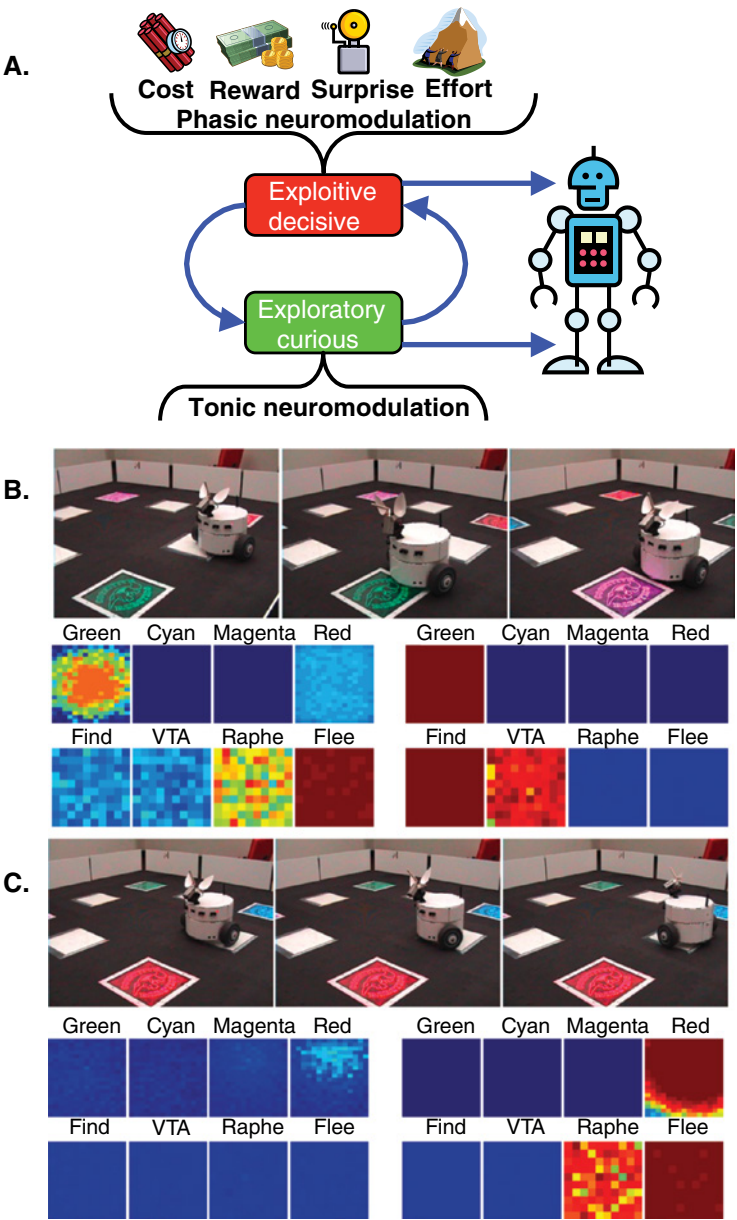
Neuromodulation and Decisiveness

In another interpretation of the empirical data, a framework was presented in which all neuromodulators to some degree control the exploration/exploitation tradeoff, but the environmental cues that trigger a neuromodulator differ (Krichmar, 2008). From the evidence, it appears that the common effect of the neuromodulatory system is to increase the signal-to-noise ratio of downstream neuronal targets such that the organism can make quick and decisive choices (see Fig. 27.8A). In this framework, the serotonergic monitored threat assessment, the cholinergic system monitored attentional effort, dopaminergic system monitored reward prediction and wanting, and the noradrenergic system monitored novelty and saliency. The framework suggested a mechanism in which during phasic neuromodulation, the system became exploitive by increasing the gain of the thalamocortical connections and the inhibitory connections in the cortex. They specifically tested this theory in a neurobotic experiment where the robot's behavior was controlled by a detailed neural network, which had thalamocortical,

corticocortical, and inhibitory connections (Cox & Krichmar, 2009). Phasic bursts of neuromodulatory activity shaped the robot's behavior and drove appropriate behavioral responses, which was to approach positive

value objects and avoid negative value objects (see Fig. 27.8B and C). The phasic response of the simulated neuromodulators caused the robot to attend to appropriate stimuli, ignore distracters, and take decisive actions.

Figure 27.8 Model of phasic neuromodulation and decisiveness. (A) In this framework, phasic neuromodulation drives the agent toward more exploitive and decisive behavior, and tonic neuromodulation drives the agent toward more exploratory or curious behavior. From Krichmar (2008). (B and C) Neurorobot model showing how phasic neuromodulation can lead to rapid and appropriate decision making, and an increase in SNR in the presence of noisy sensors. (B) Robot response and neural activity to a positive valence object. (C) Robot response and neural activity to a negative valence object. From Cox and Krichmar (2009).



Conclusions

In the present chapter, we reviewed the experimental evidence on how different neuromodulators can influence cognitive behavior. From this, we presented a number of models of neuromodulation that have influenced our thinking on how neuromodulation contributes to cognitive function. Despite neuromodulation being an active area of research, we believe more future

models should take into consideration the interaction between these neuromodulatory systems. There is much evidence supporting the notion that these systems do not work in isolation. Because of the technical difficulties examining multiple brain areas simultaneously, computational modeling has an important advantage in putting forward new holistic ideas on the function of neuromodulatory systems and their role in cognition.

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